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Batch no-074

Topic- linux basics and softlink and hardlink

- **What is linux?**

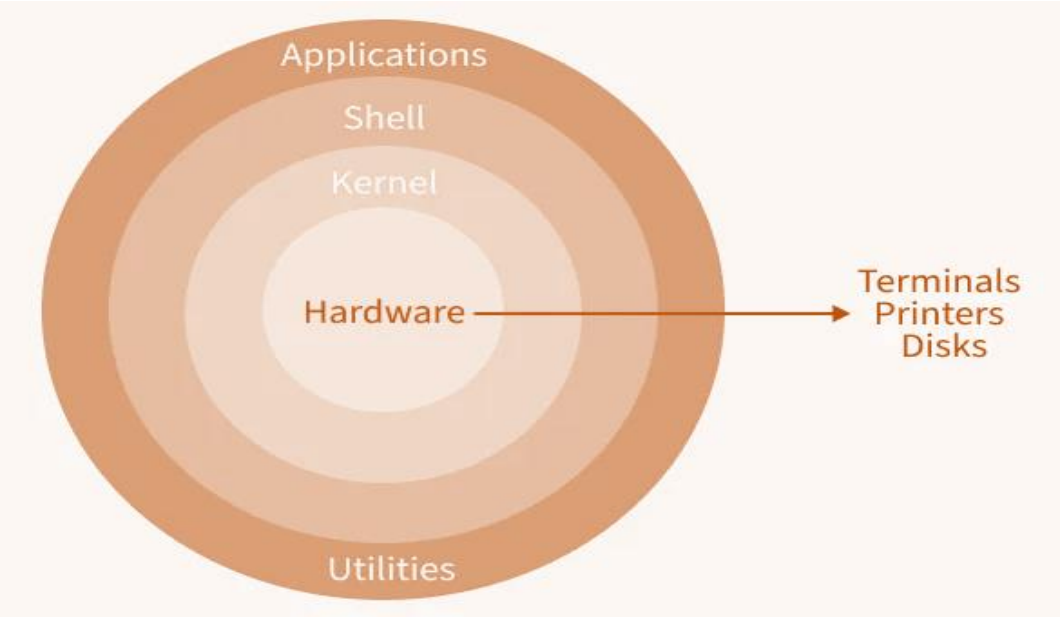
Linux is open-source operating system which is widely used in various organisation. It is free to use, fast, highly secure and reliable.

Linux Popular distributions are ubuntu, amazon linux, fedor.

Linux mainly used for programming and software development, cloud, servers, cybersecurity.

- **Architecture of linux**

Linux is made up of karnel shells and users



Linux kernel is the heart of linux it has direct access to hardware

Shell is an interface which communicates between user and system or hardware means when user gives commands to shell it understand them and ask kernel to execute it .

- **Basic linux commands**

Basic linux commands are: ls used to display contents in directory or file

1.pwd stands for present working directory

2.whoami means I'm in which folder or directory

3.cd stands for change directory.

4.mkdir stands for making new directory.

5.psudo su- is used to go into ROOT (admin level command)

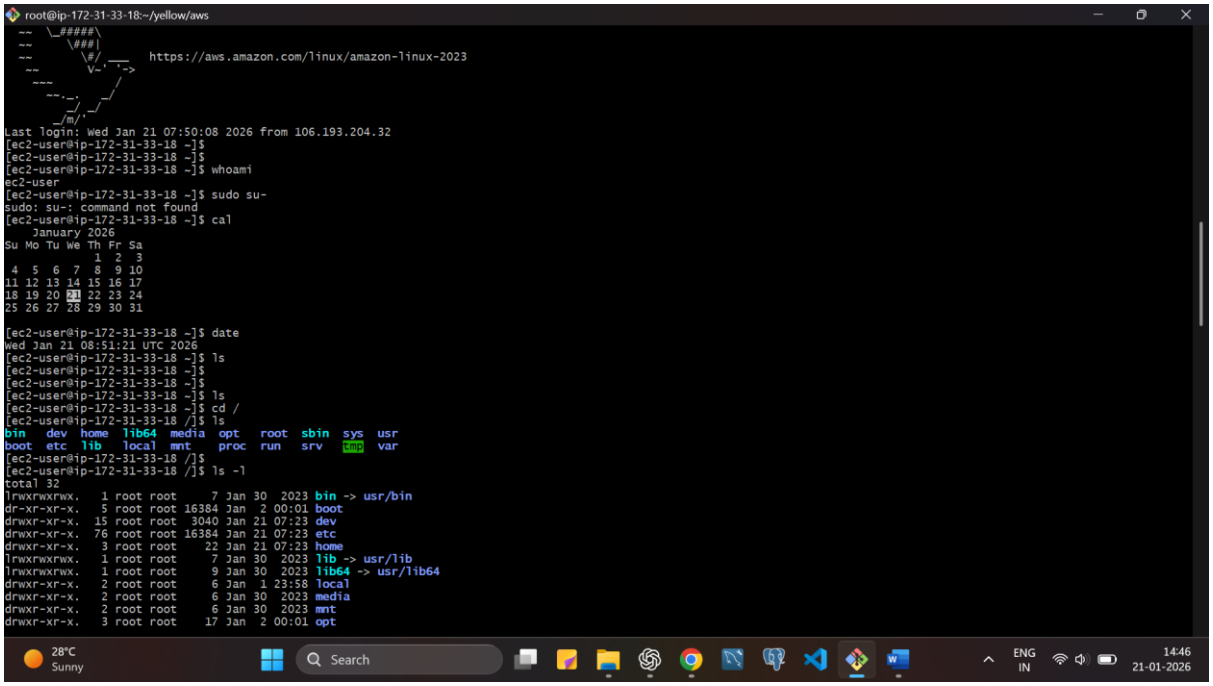
6.CAT used to read file .

7.touch is used to make empty directories.

8.cp & mv command used to copy and move files or directories

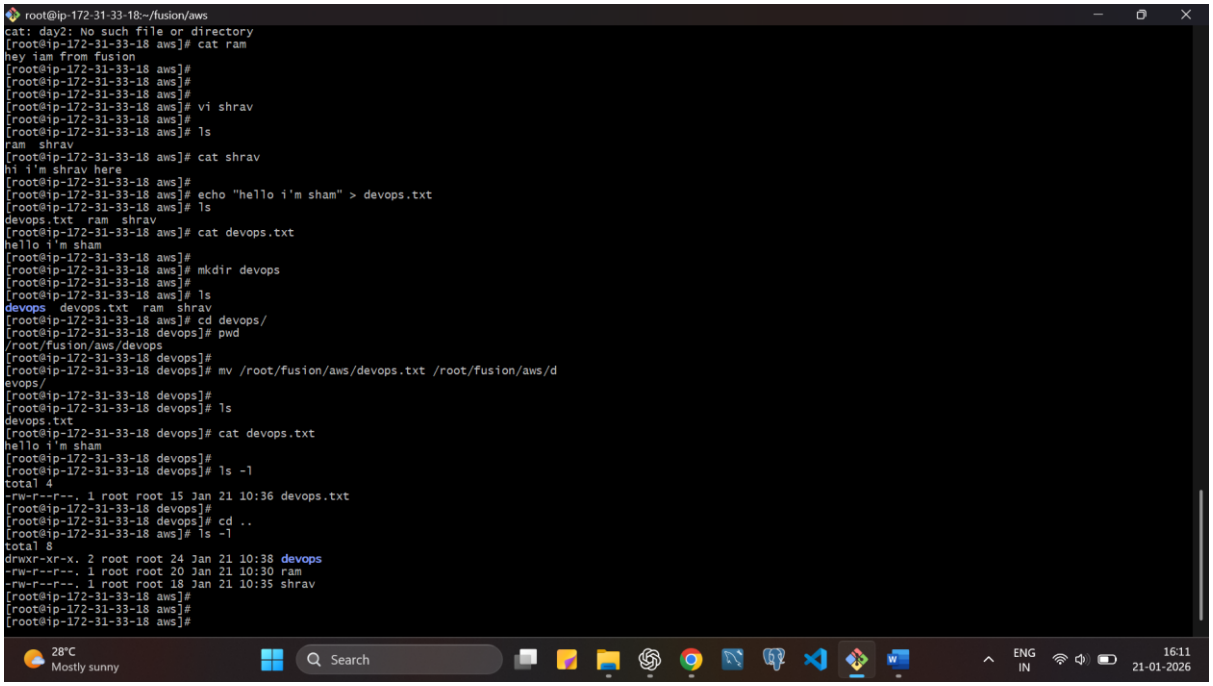
9.ls -l is used to see long list of directories.

10.date and cal use to see date and calender



```
root@ip-172-31-33-18:~/fusion/aws
[root@ip-172-31-33-18 fusion]# mkdir aws
[root@ip-172-31-33-18 fusion]# ls
aws  day1  day2  day3  day4
[root@ip-172-31-33-18 fusion]# ls -l
total 4
drwxr-xr-x. 2 root root  6 Jan 21 10:23 aws
-rw-r--r--. 1 root root 17 Jan 21 09:25 day1
-rw-r--r--. 1 root root  0 Jan 21 09:25 day2
-rw-r--r--. 1 root root  0 Jan 21 09:25 day3
-rw-r--r--. 1 root root  0 Jan 21 09:25 day4
[root@ip-172-31-33-18 fusion]#
[root@ip-172-31-33-18 fusion]#
[root@ip-172-31-33-18 fusion]# cd aws
[root@ip-172-31-33-18 aws]# pwd
/root/fusion/aws
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# cp /fusion/day1 /fusion/aws/ram
cp: cannot stat '/fusion/day1': No such file or directory
[root@ip-172-31-33-18 aws]# ls
[root@ip-172-31-33-18 aws]# cat ram
cat: ram: No such file or directory
[root@ip-172-31-33-18 aws]# cp
cp: missing file operand
Try 'cp --help' for more information.
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# cd..
-bash: cd..: command not found
[root@ip-172-31-33-18 aws]# cd ..
[root@ip-172-31-33-18 fusion]# ls
aws  day1  day2  day3  day4
[root@ip-172-31-33-18 fusion]# ls -l
total 4
drwxr-xr-x. 2 root root  6 Jan 21 10:23 aws
-rw-r--r--. 1 root root 17 Jan 21 09:25 day1
-rw-r--r--. 1 root root  0 Jan 21 09:25 day2
-rw-r--r--. 1 root root  0 Jan 21 09:25 day3
-rw-r--r--. 1 root root  0 Jan 21 09:25 day4
[root@ip-172-31-33-18 fusion]#
[root@ip-172-31-33-18 fusion]# echo "hey iam from fusion" > day2
[root@ip-172-31-33-18 fusion]# ls -l
total 8
drwxr-xr-x. 2 root root  6 Jan 21 10:23 aws
-rw-r--r--. 1 root root 17 Jan 21 09:25 day1
-rw-r--r--. 1 root root 20 Jan 21 10:27 day2
-rw-r--r--. 1 root root  0 Jan 21 09:25 day3
-rw-r--r--. 1 root root  0 Jan 21 09:25 day4
[root@ip-172-31-33-18 fusion]# cat day3
[root@ip-172-31-33-18 fusion]# cat day2
```

```
root@ip-172-31-33-18:~/fusion/aws
[root@ip-172-31-33-18 fusion]# ls -l
total 8
drwxr-xr-x. 2 root root  6 Jan 21 10:23 aws
-rw-r--r--. 1 root root 17 Jan 21 09:25 day1
-rw-r--r--. 1 root root 20 Jan 21 10:27 day2
-rw-r--r--. 1 root root  0 Jan 21 09:25 day3
-rw-r--r--. 1 root root  0 Jan 21 09:25 day4
[root@ip-172-31-33-18 fusion]# cat day3
[root@ip-172-31-33-18 fusion]# cat day2
hey iam from fusion
[root@ip-172-31-33-18 fusion]#
[root@ip-172-31-33-18 fusion]#
[root@ip-172-31-33-18 fusion]# cp /fusion/day2 /fusion/aws/
cp: cannot stat '/fusion/day2': No such file or directory
[root@ip-172-31-33-18 fusion]# ^C
[root@ip-172-31-33-18 fusion]# cp /root/fusion/day2 /root/fusion/aws/ram
[root@ip-172-31-33-18 fusion]#
[root@ip-172-31-33-18 fusion]# ls
aws  day1  day2  day3  day4
[root@ip-172-31-33-18 fusion]# cd aws
[root@ip-172-31-33-18 aws]# ls -l
total 4
-rw-r--r--. 1 root root 20 Jan 21 10:30 ram
[root@ip-172-31-33-18 aws]# cat day2
cat: day2: No such file or directory
[root@ip-172-31-33-18 aws]# cat ram
hey iam from fusion
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# vi shrav
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# ls
ram  shrav
[root@ip-172-31-33-18 aws]# cat shrav
hi i'm shrav here
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# echo "hello i'm sham" > devops.txt
[root@ip-172-31-33-18 aws]# ls
devops.txt  ram  shrav
[root@ip-172-31-33-18 aws]# cat devops.txt
hello i'm sham
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# mkdir devops
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# ls
devops  devops.txt  ram  shrav
[root@ip-172-31-33-18 aws]# cd devops/
```



```
root@ip-172-31-33-18:~/fusion/aws
cat: day2: No such file or directory
[root@ip-172-31-33-18 aws]# cat ram
hey iam from fusion
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# vi shrav
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# ls
ram  shrav
[root@ip-172-31-33-18 aws]# cat shrav
hi i'm shrav here
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# echo "hello i'm sham" > devops.txt
[root@ip-172-31-33-18 aws]# ls
devops.txt  ram  shrav
[root@ip-172-31-33-18 aws]# cat devops.txt
hello i'm sham
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# mkdir devops
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]# ls
devops  devops.txt  ram  shrav
[root@ip-172-31-33-18 aws]# cd devops/
[root@ip-172-31-33-18 devops]# pwd
/root/fusion/aws/devops
[root@ip-172-31-33-18 devops]#
[root@ip-172-31-33-18 devops]# mv /root/fusion/aws/devops.txt /root/fusion/aws/d
evops/
[root@ip-172-31-33-18 devops]#
[root@ip-172-31-33-18 devops]# ls
devops.txt
[root@ip-172-31-33-18 devops]# cat devops.txt
hello i'm sham
[root@ip-172-31-33-18 devops]#
[root@ip-172-31-33-18 devops]# ls -l
total 4
-rw-r--r--. 1 root root 15 Jan 21 10:36 devops.txt
[root@ip-172-31-33-18 devops]#
[root@ip-172-31-33-18 devops]# cd ..
[root@ip-172-31-33-18 aws]# ls -l
total 8
drwxr-xr-x. 2 root root 24 Jan 21 10:38 devops
-rw-r--r--. 1 root root 20 Jan 21 10:30 ram
-rw-r--r--. 1 root root 18 Jan 21 10:35 shrav
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]#
[root@ip-172-31-33-18 aws]#
```

- **What is soft link?**

Soft link is type of file which link to other file by its path or directory. This only stores the path not the actual data without using extra space

This is use to create a shortcuts.

Example : shortcuts in windows,application on home screen of an android mobile

- **What is hard link?**

A hard link is another name for the same file.

Both files point to the same data on disk, use the same storage space, and behave like original files.

In this data is removed only if all hardlinks are deleted

This act as a backup system for sensitive data .